

Test an API Using Swagger UI

Hands-on Walkthrough using the Swagger Petstore Demo

Hands-On	16
Scenario	Test API Using Swagger UI
Tool	Swagger UI (browser)
Demo Site	https://petstore.swagger.io/
Sample Endpoint	GET /pet/findByStatus
Auth	None (public demo)
Duration	30 minutes
Prerequisite	Modern web browser + internet connection

Objective

The objective of this assignment is to test an API directly from a web browser using **Swagger UI**. Swagger UI is an interactive web interface that renders an OpenAPI specification as a clickable, executable API documentation page. The official **Swagger Petstore demo** is used as the sandbox — it exposes a small set of pet, store, and user endpoints that anyone can call without authentication.

What is Swagger UI?

Swagger UI is the most widely-used renderer for the OpenAPI (Swagger) specification. It generates an interactive page that shows every endpoint in an API along with its method, parameters, request body schema, and possible responses. Each endpoint can be executed live from the browser using a built-in **Try it out** feature — no Postman, curl, or code is required. Behind the scenes, Swagger UI builds a regular HTTP request from the inputs and shows the response on the same page.

Note — HTTP Method Colors in Swagger UI

Swagger UI follows a colour convention so the method of every endpoint can be identified at a glance: **GET** is green, **POST** is blue, **PUT** is orange, **DELETE** is red, and **PATCH** is teal. The colour is shown both as a tag next to the path and as a coloured left-hand bar when the endpoint is expanded.

Steps

1 Open the Swagger Petstore in a Browser

Open a modern web browser (Chrome, Edge, Firefox, Safari).

Navigate to the URL:

<https://petstore.swagger.io/>

Swagger UI loads with the **Swagger Petstore** documentation. The endpoints are grouped into three tags: **pet**, **store**, and **user**.

2 Explore the API Documentation

Scroll down to see the list of endpoints under each tag. Each row shows the HTTP method (color-coded), the path, and a short description.

Click any endpoint row to expand it and reveal the full details: parameters, request body schema, possible response codes, and a sample response.

This documentation is auto-generated from the API's OpenAPI spec at `/v2/swagger.json` — visible in the bar at the top of the page.

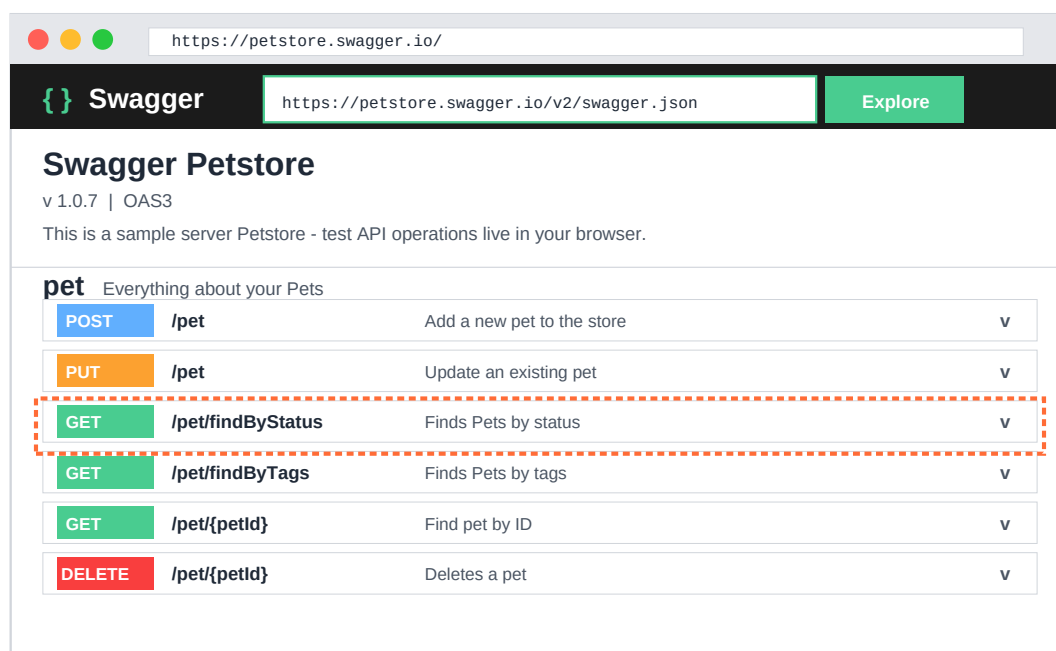


Figure 1 — Swagger Petstore homepage showing the list of endpoints under the **pet** tag, color-coded by HTTP method. **GET /pet/findByStatus** is highlighted as the endpoint chosen for this exercise.

3 Select an API Endpoint

Choose the endpoint to test. For this walkthrough, use:

GET /pet/findByStatus — this returns a list of pets filtered by their status. It needs no auth and returns a JSON array.

Other simple endpoints to try later: **GET /pet/{petId}** (fetch a single pet by id), **GET /store/inventory** (counts of pets per status), and **POST /pet** (add a new pet).

4 Click on the Endpoint to Expand It

Click anywhere on the **GET /pet/findByStatus** row.

The row expands downward to reveal the full endpoint contract:

- A **Parameters** section listing every field the endpoint accepts.
- A **Responses** section listing every possible response code with sample bodies and schemas.
- A **Try it out** button on the right that switches Parameters into editable mode.

5 Fill In the Request Parameters

Click the **Try it out** button on the right side of the Parameters block. Postman-style input boxes appear next to each parameter.

For **GET /pet/findByStatus**, the only required parameter is **status** (an array of strings). The dropdown offers three valid values: **available**, **pending**, and **sold**.

Choose **available** from the dropdown. (Multiple values can be selected to filter on more than one status.)

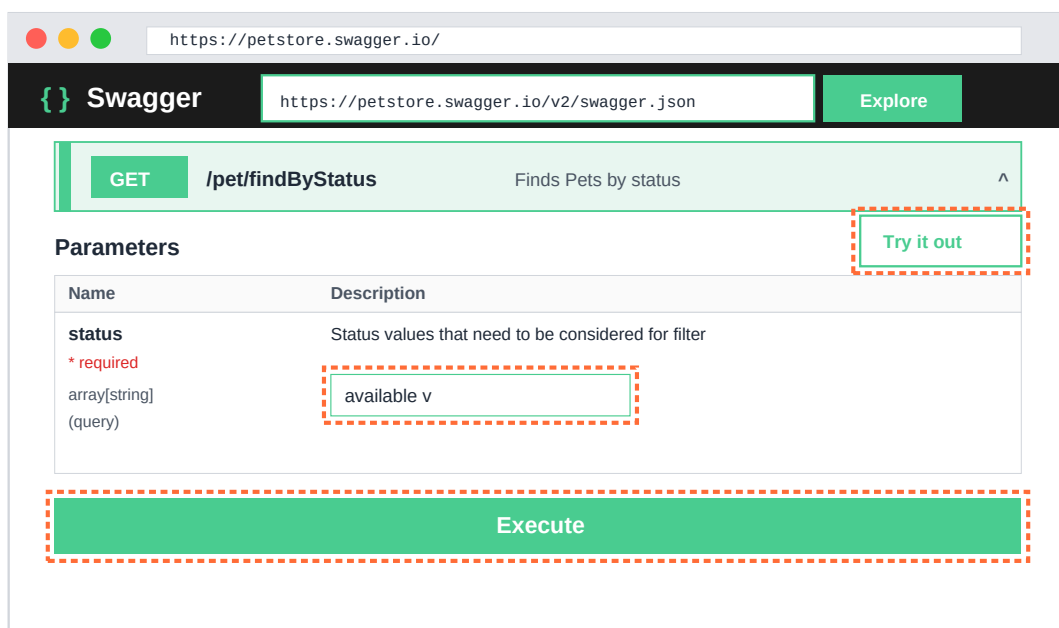


Figure 2 — Endpoint expanded with method tag **GET**, the **status** required parameter highlighted with the dropdown set to **available**, and the **Execute** button highlighted at the bottom.

6 Add Headers / Authentication If Required

Some endpoints require an API key, OAuth token, or basic auth. These are configured globally via the **Authorize** button at the top-right of the page (a padlock icon).

GET /pet/findByStatus on the public Petstore demo is open, so no headers or credentials are needed.

For endpoints that *do* require an `api_key` (such as **DELETE /pet/{petId}**), click **Authorize**, paste a value into the `api_key` field, and click **Authorize** again. Swagger UI will attach the header automatically to every subsequent request.

7 Click "Try it out" / "Execute" to Send

After clicking **Try it out** in step 5, the green **Execute** button appears at the bottom of the parameters block.

Click **Execute**. Swagger UI builds the HTTP request, sends it to the API, and renders the response below.

While the call is in flight, the button briefly shows a loading spinner.

8 Inspect the Response

Swagger UI displays four pieces of information in the Responses section after a successful call:

- **Curl** — the equivalent `curl` command, useful for copying into a terminal.
- **Request URL** — the exact URL that was hit, including any query string.
- **Server response** — the HTTP status code (200 OK for a successful GET) and the full response body in pretty-printed JSON.
- **Response headers** — if the API returns any.

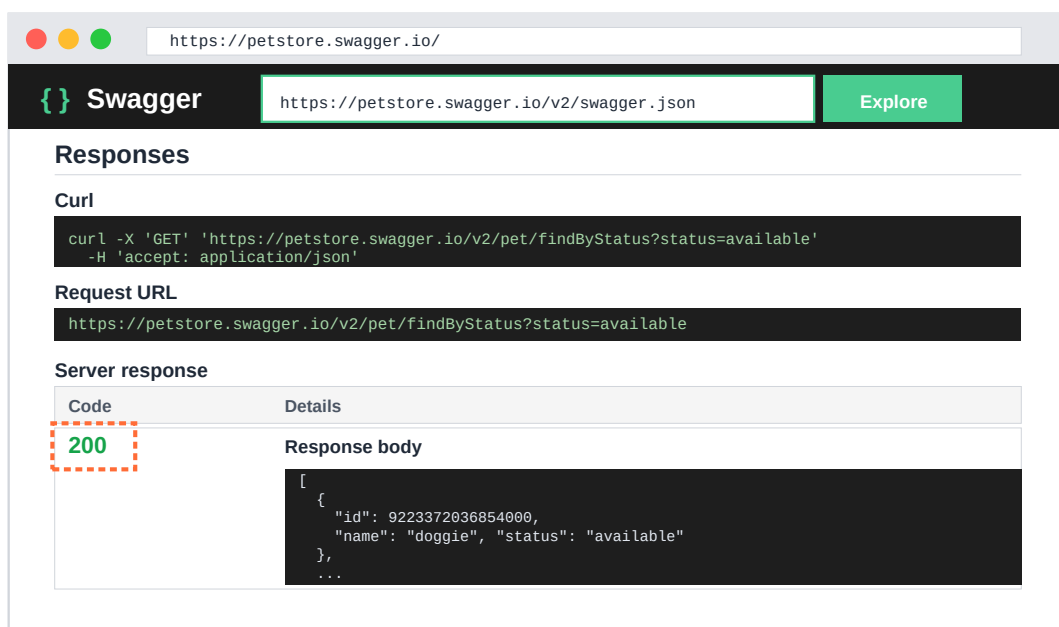


Figure 3 — Response section showing the equivalent **curl** command, the resolved **Request URL**, a **200** status code (highlighted), and a sample of the JSON **Response body** returned by the server.

9 Verify the Data Returned

Examine the response body and confirm:

- The HTTP status code is **200** (or whatever the endpoint documents as the success code).
- The body is a non-empty JSON array.
- Every pet object in the array has a **status** equal to the value sent (*available* in this case).
- Required fields like **id** and **name** are present on every record.

If any of these are wrong (e.g. status mismatch, missing fields, or a 4xx / 5xx code), this indicates a defect in the API or in the request being sent. Capture the curl command and request URL so the issue can be reproduced and reported.

Sample Response Body

The response is a JSON array; each element is a pet object. A shortened example:

```
[
  {
    "id": 9223372036854000,
    "category": { "id": 1, "name": "Dogs" },
    "name": "doggie",
    "photoUrls": [ "string" ],
    "tags": [ { "id": 0, "name": "string" } ],
    "status": "available"
  },
  { ... },
  { ... }
]
```

Because Petstore is a shared sandbox used by many people, the exact contents of the array change continuously. The **shape** of each item should match the schema in the documentation, even if the values vary.

Verification Checklist

- 1 The browser successfully reaches <https://petstore.swagger.io/> and renders the Swagger UI page.
- 2 The **pet**, **store**, and **user** sections are visible with their endpoints listed.
- 3 The chosen endpoint expands cleanly and shows its Parameters and Responses sections.
- 4 After clicking **Try it out** and **Execute**, the page shows a **200** status code in green under Server response.
- 5 The response body is a JSON array of pet objects, each with **id**, **name**, and **status** fields.
- 6 Every returned pet has **status** matching the value selected in the parameters.

Conclusion

An API endpoint has been successfully tested directly from the browser using Swagger UI. The same workflow — expand the endpoint, click Try it out, fill in parameters, click Execute, verify the response — applies to any OpenAPI-documented API. Swagger UI is a fast way to explore unfamiliar APIs,

sanity-check expected behaviour, and gather example payloads that can later be imported into Postman or REST Assured for more thorough automated testing.